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[Google Scholar](#) | [ORCID](#) | Dhaka, Bangladesh

RESEARCH SUMMARY

Focused on mathematically grounded, efficient transformer inference and long-context optimization; targeting Fall 2027 PhD admissions.

RESEARCH PROJECTS

Mathematical Foundations of Efficient Transformers

- Developing a theoretical framework for transformer inference, emphasizing KV cache compression and attention scaling.
- Curating rigorous derivations in linear algebra and optimization to advance doctoral research on LLM efficiency.

EXPERIENCE

North South University

May 2019 – May 2025

Lab Instructor

Dhaka, Bangladesh

- Instructed 1000+ undergraduates in **core CS theory** and algorithmic problem-solving; led curriculum design and assessment strategies.

Me-Solshare Ltd

Nov. 2019 – Aug. 2021

Jr. Data Scientist

Dhaka, Bangladesh

- Engineered **high-accuracy CNNs** for time-series energy forecasting, utilizing rigorous statistical evaluation methods.

Power Ledger Pty Ltd

Oct. 2022 – Feb. 2024

Product Analyst

Remote

- Managed **technical data analysis and QA** for a blockchain platform, facilitating cross-functional alignment.

PUBLICATIONS & PROJECTS

End-to-End OCR via Synthetic Dataset Generation (*IJCCI 2020, Springer*)

- Trained a **deep-learning OCR model** on synthetically generated noisy images; improved robustness in low-quality document recognition | [doi:10.1007/978-981-15-3607-6_41](https://doi.org/10.1007/978-981-15-3607-6_41)

Population Estimation of Rohingya Refugees via ANN (*VJCS 2019, World Scientific*)

- Modeled refugee population using **Artificial Neural Networks** on satellite and census data; applied to humanitarian resource planning | [doi:10.1142/S2196888819500246](https://doi.org/10.1142/S2196888819500246)

Early Detection of Glaucoma Using Fuzzy Logic in Bangladesh Context (*IS 2018, IEEE*)

- Built an **ANFIS-based fuzzy logic model** for early glaucoma detection from OCT and tonometry data; addressed diagnostic access gaps in Bangladesh | [doi:10.1109/IS.2018.8710490](https://doi.org/10.1109/IS.2018.8710490)

CarbonIQ — AI-Powered Carbon Accounting Platform

- Designed **time-series CO₂e forecasting pipelines** and automated ETL/ELT workflows; built multi-tenant API architecture with containerized, reproducible MLOps infrastructure

EDUCATION

North South University

2015 – 2018

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

- Honors:** CGPA 3.62/4.00. **Core:** ML, Algorithms, Databases, Systems, OOP.
- Merit Scholarship:** 25% tuition scholarship awarded based on academic performance.
- Certifications:** Deep Learning Specialization (DeepLearning.AI, 2022), Machine Learning Specialization (Stanford/Coursera, 2021).

SKILLS

Languages: Python, C/C++, Java, JS/TS, SQL

ML & Research: PyTorch, TensorFlow, NumPy/Pandas, SciKit-Learn, MLflow

Tools & Cloud: FastAPI, Docker, PostgreSQL/MongoDB, AWS, Linux, $\LaTeX$